

Ê PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

k_i = \frac{\omega \sqrt{\epsilon_{r,i}}}{c} (rad/m) \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}

Direction table u \in \{x, y, z\} = axis perp. to boundary

Inc dir	Inc	Refl	Trans
+û	e^{-jk_1 u}	e^{+jk_1 u}	e^{-jk_2 u}
-û	e^{+jk_1 u}	e^{-jk_1 u}	e^{+jk_2 u}

+û direction \Leftrightarrow Med 2 at u \ge 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a_0):

E^i = a_0 ê e^{\mp j k_1 u}

E^r = \Gamma a_0 ê e^{\pm j k_1 u}

E^t = \tau a_0 ê e^{\mp j k_2 u}

(lossy med 2: replace e^{\mp j k_2 u} with e^{\mp \alpha_2 u} e^{\mp j \beta_2 u})

Total in medium 1: E_1 = E^i + E^r.

ê by polarization *plug into templates above*

Pol.	ê	Note
Lin. x̂	x̂	E_y = 0
Lin. ŷ	ŷ	E_x = 0
Lin. at 45°	(x̂ + ŷ)/√2	in phase
RHC	x̂ - j ŷ	δ = -π/2
LHC	x̂ + j ŷ	δ = +π/2
General	a_x x̂ + a_y e^{jδ} ŷ	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

%P_r = 100 |Γ|^2

% transmitted *η-ratio matters!*

%P_t = 100 |τ|^2 \frac{\eta_1}{\eta_2}

Check: %P_r + %P_t = 100.

If medium 2 is denser (η_2 < η_1), |τ| > 1 but %P_t < 100% because of the η_1/η_2 factor.

LOSSY MEDIA — 5 CASES

Loss tangent ε' = ε_r ε_0, ε_0 ≈ 1/(36π × 10^9)

\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}

Type	σ/ωε	ηc, α, β
Lossless	0	η = η_0/√ε_r exact; α = 0, β = ω√ε_r/c
Low-loss	< 10^-2	\frac{\eta_0}{\sqrt{\epsilon_r}} \approx \frac{\eta_0}{\sqrt{\epsilon_r}} (1 + j \frac{\sigma}{2\omega \epsilon'}) ; \alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}
Quasi-cond.	10^-2 - 10^2	exact: (η/√X) e^{jθ} η, see below
Good cond.	> 10^2	η_c ≈ (1+j)√π f μ σ; α ≈ β ≈ √π f μ σ
Magnetic any		keep full √μ_r/ε_r η_0

For Γ/τ: drop j correction; use η_0/√ε_r. Magnetic: keep μ_r.

Exact (quasi-cond) X = √(1 + (ε''/ε')^2)

\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X - 1}

\beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X + 1}

\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}

BOUNDARY

η_1, η_2 — imp. (Ω)
Γ — refl. coeff (unitless)
t_i = 1 + Γ — trans. coeff
k_i = ω√ε_{r,i}/c (rad/m)
α_2, β_2 — if med 2 lossy
z = 0 — boundary plane
η_0 = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k̂+normal
⊥: E ⊥ this plane
||: E || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
— prop. const (rad/m)
Z_{TE/TM} — wave imp. (Ω)
E_x, H_y — field comps (V/m, A/m)
η = η_0/√ε_r

VELOCITIES

u_{p0} = c/√ε_r — bulk phase vel.
u_p = u_{p0}/√(1 - (f_c/f)^2)
u_g = u_{p0}/√(1 - (f_c/f)^2)
u_p u_g = u_{p0}^2
t = L/u_g — travel time
c = 3 × 10^8 m/s

POWER & UNITS

%P_r = 100 |Γ|^2
%P_t = 100 |τ|^2 η_1/η_2
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε_0 = 8.85 × 10^-12 F/m
μ_0 = 4π × 10^-7 H/m
1/(36π) × 10^-9 ≈ ε_0

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

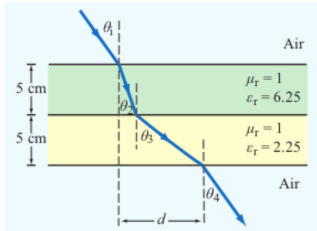
\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}

Air-slabs-air shortcut *outer media identical*

\theta_{\text{exit}} = \theta_{\text{incident}}

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i

Slab-indexed θ_i = angle inside slab i

d = \sum_{i=1}^N t_i \tan \theta_i

Angle-indexed (HW3 P3 style) θ_1 = incident, θ_{i+1} = inside slab i

d = \sum_{i=1}^N t_i \tan \theta_{i+1}

Same physics, different numbering. In HW3 P3: slab 1 contributes t_1 tan θ_2, slab 2 contributes t_2 tan θ_3. The angle in the formula is always the one the ray makes inside that slab.

Air-slab-air: exit angle returns to θ_1 but is laterally shifted by d.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted. Shorthand: c_i = cos θ_i, c_t = cos θ_t.

	Normal (θ_i = 0)	⊥ pol	pol
Γ	\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}	\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}	\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}
τ	\frac{2\eta_2}{\eta_2 + \eta_1}	\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}	\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}
τ rel.	τ = 1 + Γ	τ_⊥ = 1 + Γ_⊥	τ_ = (1 + Γ_) \frac{c_i}{c_t}
R	Γ ^2	Γ_⊥ ^2	Γ_ ^2
T	\tau ^2 \frac{\eta_1}{\eta_2}	\tau_⊥ ^2 \frac{\eta_1 c_t}{\eta_2 c_i}	\tau_ ^2 \frac{\eta_1 c_t}{\eta_2 c_i}
R+T	T = 1 - R	T_⊥ = 1 - R_⊥	T_ = 1 - R_

Notes: sin θ_t = √(μ_1 ε_1/μ_2 ε_2) sin θ_i; nonmag. ⇒ η_2/η_1 = n_1/n_2.

Intrinsic impedance η_i *plug into table above*

\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\epsilon_{r,i}} (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega

Default μ_r = 1 (air, dielectric). Use μ_r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction η_c σ/(ωε) ≪ 1

\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega \epsilon'} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}

For Γ/τ just use η_0/√ε_r. The j term is for |η_c|, θ_η. See LOSSY MEDIA — 5 CASES for all regimes.

RECTANGULAR WAVEGUIDE — MODES

TE mode E_z = 0, H_z ≠ 0

TM mode H_z = 0, E_z ≠ 0

Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.

E_x form (TE) *read m, n from arguments*

E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)

coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{mn} vs TM_{mn} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).

3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. That's why TE_{10} exists but TM_{10} doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: C_X = cos \frac{m\pi x}{a}, S_X = sin \frac{m\pi x}{a}, C_Y = cos \frac{n\pi y}{b}, S_Y = sin \frac{n\pi y}{b}. All fields carry an e^{-j\beta z} factor (omitted below). k_c^2 = (m\pi/a)^2 + (n\pi/b)^2.

TE mode *generator: H̃_z*

\tilde{H}_z = H_0 C_X C_Y

\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y

\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y

\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TE}

TM mode *generator: Ẽ_z*

\tilde{E}_z = E_0 S_X S_Y

\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y

\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y

\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TM}

TEM (plane wave) for reference: Ẽ_x = E_0 e^{-jβz}, H̃_y = Ẽ_x/η, f_c N/A, k = ω√μ̵.

CUTOFF FREQUENCY

f_{mn} *m, n integers ≥ 0 (not both zero)*

f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}

Common cutoffs (a > b)

Quantity	Formula
u_{p0}	c/√ε_r (hollow: u_{p0} = c)
TE_{10}	u_{p0}/(2a) (dominant)
TE_{01}	u_{p0}/(2b)
TE_{20}	u_{p0}/a = 2f_{10}
TE_{11}, TM_{11}	\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}

Propagation requires f > f_{mn}.

ε_r FROM DISPERSION

Given ω, β, mode (m, n)

\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} (\Omega) \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}

\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}

Z_{TE} > η always (denominator < 1); Z_{TM} < η.

GROUP VELOCITY & TRAVEL TIME

u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} (m/s)
u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} (m/s) ⇒ u_p u_g = u_{p0}^2 (m^2/s^2)
t = L/u_g (s) *travel time through guide of length L*
Higher modes have lower u_g ⇒ they arrive later. Group delay spread = sum over excited modes.